CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 12: Algebraic Identities

Every formula and result of the chapter on one sheet

1	Square of a sum	$(a+b)^2 = a^2 + 2ab + b^2$
2	Square of a difference	$(a-b)^2 = a^2 - 2ab + b^2$
3	Difference of squares	$(a+b)(a-b) = a^2 - b^2$
4	Product	$(x+a)(x+b) = x^2 + (a+b)x + ab$
5	Sum of the squares	$(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$
6	Difference of the squares	$(a+b)^2 - (a-b)^2 = 4ab$
7	Sum of squares	$a^2 + b^2 = (a+b)^2 - 2ab$
8	Reciprocal form	$x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$
9	Trinomial square	$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$
10	Cube of a sum	$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
11	Cube of a difference	$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
12	Sum of cubes	$a^3 + b^3 = (a+b)^3 - 3ab(a+b)$